3.4 泰勒公式

1 两种余项的泰勒公式

2 常见函数的泰勒公式

3 泰勒公式的应用

一、问题的提出

1. 设 $f(x)$在x_0处可导, 则有

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

$$[f(x) = f(x_0) + f'(x_0)(x - x_0) + o(x - x_0)]$$

例如, 当$|x|$很小时, $e^x \approx 1 + x$,

$$\ln(1 + x) \approx x$$

不足: 1、精确度不高；　2、误差不能估计.

问题: 寻找函数$P(x)$, 使得 $f(x) \approx P(x)$

误差 $R(x) = f(x) - P(x)$ 可估计

令$P(x)$是一个关于$x - x_0$的多项式函数

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots + a_n(x - x_0)^n$$

误差 $R_n(x) = f(x) - P_n(x)$

二、P_n和R_n的确定

分析: 1.在 x_0点相交

$$P_n(x_0) = f(x_0)$$

2.若有相同的切线

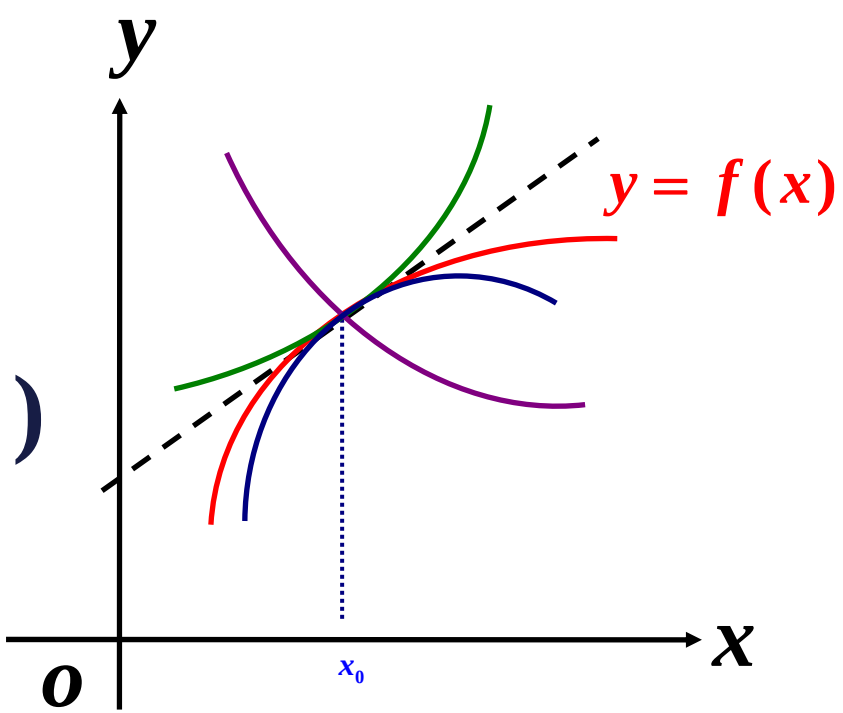
$$P_n'(x_0) = f'(x_0)$$

3.若弯曲方向相同

$$P_n''(x_0) = f''(x_0)$$

…… ……

近似程度越来越好



假设 $P_n^{(k)}(x_0) = f^{(k)}(x_0) \quad k = 1,2,\cdots,n$

$$a_0 = f(x_0),$$
$$1 \cdot a_1 = f'(x_0),$$
$$2! \cdot a_2 = f''(x_0)$$
$$\cdots \quad \cdots,$$
$$n! \cdot a_n = f^{(n)}(x_0)$$

$$\therefore a_k = \frac{1}{k!} f^{(k)}(x_0)$$

$$(k = 0,1,2,\cdots,n)$$

代入$P_n(x)$中得

$$P_n(x)=f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2+$$

$$\cdots+\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

$P_n(x)$称为$f(x)$在x_0点的n阶泰勒多项式。

三、泰勒(Taylor)中值定理

定理8　设函数$f(x)$在点x_0处具有n阶的导数，

则存在x_0点的邻域(a,b)，对$\forall x \in (a,b)$，有

$$f(x)=f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2$$

$$+\cdots+\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n+\underline{o[(x-x_0)^n]}$$

皮亚诺形式的余项

定理9　设函数$f(x)$在包含x_0的邻域(a,b)内

具有直到$n+1$阶的导数，则对 $\forall x \in (a,b)$,

存在介于x与x_0之间的ξ，使得

$$f(x)=f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2$$

$$+\cdots+\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n+\underline{\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}}$$

$$P_n(x)=\sum_{k=0}^{n}\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

称为 $f(x)$ 按 $(x-x_0)$ 的幂展开的 n 次近似多项式

$$f(x)=\sum_{k=0}^{n}\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k+R_n(x)$$

称为 $f(x)$ 按 $(x-x_0)$ 的幂展开的 n 阶泰勒公式

$$R_n(x)=\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}\quad(\xi\text{在}x_0\text{与}x\text{之间})$$

拉格朗日形式的余项

注意:

1. 当$n=0$时, 泰勒公式变成拉氏中值公式

$$f(x)=f(x_0)+f'(\xi)(x-x_0)\quad(\xi\text{在}x_0\text{与}x\text{之间})$$

2. ξ介于x与x_0之间，余项

$$R_n(x)=\frac{f^{(n+1)}(x_0+\theta(x-x_0))}{(n+1)!}(x-x_0)^{n+1},(0<\theta<1)$$

麦克劳林(Maclaurin)公式

$$f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\cdots+\frac{f^{(n)}(0)}{n!}x^n$$

$$+\frac{f^{(n+1)}(\theta x)}{(n+1)!}x^{n+1} \qquad (0<\theta<1)$$

$$f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\cdots+\frac{f^{(n)}(0)}{n!}x^n$$

$$+o(x^n)$$

例1. 求f(x)=tanx在x=π/4处的3 阶泰勒公式。

解: $f(x)=\tan x,\quad f'(x)=\sec^2 x,$

$$f''(x)=2\sec^2 x\tan x=2(\tan^3 x+\tan x),$$

$$f'''(x)=6\tan^2 x\sec^2 x+2\sec^2 x=6\tan^4 x+8\tan^2 x+2,$$

$$f^{(4)}(x)=24\tan^3 x\sec^2 x+16\tan x\sec^2 x$$

$$=8\sec^2 x\tan x(3\tan^2 x+2),$$

$$f(\frac{\pi}{4})=1,\ f'(\frac{\pi}{4})=2,\ f''(\frac{\pi}{4})=4,\quad f'''(\frac{\pi}{4})=16,$$

$$\therefore\ f(x)=\tan x=1+2(x-\frac{\pi}{4})+\frac{4}{2!}(x-\frac{\pi}{4})^2+\frac{16}{3!}(x-\frac{\pi}{4})^3$$

$$+\frac{8}{4!}\sec^2\xi\tan\xi(3\tan^2\xi+2)(x-\frac{\pi}{4})^4,\quad \xi\text{介于}x\text{与}\frac{\pi}{4}\text{之间}.$$

例2. 求f(x)=tanx在x=0处的3 阶麦克劳林公式.

解: $f(0)=0,\quad f'(0)=1,\quad f''(0)=0,\quad f'''(0)=2,$

$$f(x)=\tan x=0+x+\frac{0}{2!}x^2+\frac{2}{3!}x^3$$

$$+\frac{8}{4!}\sec^2\xi\tan\xi(3\tan^2\xi+2)x^4,$$

$$f(x)=\tan x=x+\frac{1}{3}x^3+\frac{8}{4!}\sec^2\xi\tan\xi(3\tan^2\xi+2)x^4,$$

其中ξ介于x与0之间.

四、简单的应用

例 1　求 $f(x)=e^x$ 的 n 阶麦克劳林公式.

解 $\because$ $f'(x)=f''(x)=\cdots=f^{(n)}(x)=e^x$,

$\therefore$ $f(0)=f'(0)=f''(0)=\cdots=f^{(n)}(0)=1$

注意到 $f^{(n+1)}(\theta x)=e^{\theta x}$ 代入公式,得

$$e^x=1+x+\frac{x^2}{2!}+\cdots+\frac{x^n}{n!}+\frac{e^{\theta x}}{(n+1)!}x^{n+1}\quad(0<\theta<1).$$

由公式可知 $e^x \approx 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}$

估计误差 (设 $x > 0$)

$$|R_n(x)| = \left| \frac{e^{\theta x}}{(n+1)!} x^{n+1} \right| < \frac{e^x}{(n+1)!} x^{n+1} (0 < \theta < 1).$$

取 $x = 1$, $e \approx 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{n!}$

其误差 $|R_n| < \frac{e}{(n+1)!} < \frac{3}{(n+1)!}$.

几个常用函数的泰勒公式

$$A:\quad f(x)=\sin x$$

$$f^{(k)}(x)=(\sin x)^{(k)}=\sin(x+\frac{k\pi}{2})\qquad f^{(k)}(0)=\sin\frac{k\pi}{2}$$

$$f^{(2k)}(0)=0\qquad f^{(2k+1)}(0)=(-1)^k\quad k=0,1,\ldots,m$$

取$n=2m$,则f(x)的$2m$阶麦克劳林公式为

$$\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}+\cdots+(-1)^{m-1}\frac{x^{2m-1}}{(2m-1)!}+\frac{\sin(\theta x+\frac{2m+1}{2}\pi)x^{2m+1}}{(2m+1)!}$$

$$(0<\theta<1)$$

B: $f(x)=\cos x$

$$f^{(k)}(x)=(\cos x)^{(k)}=\cos(x+\frac{k\pi}{2}) \qquad f^{(k)}(0)=\cos\frac{k\pi}{2}$$

$$f^{(2k)}(0)=(-1)^k \qquad f^{(2k+1)}(0)=0 \quad k=0,1,\ldots,m$$

取$n=2m+1$,则f(x)的$2m+1$阶麦克劳林公式为

$$\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}+\cdots+(-1)^m\frac{x^{2m}}{(2m)!}+\frac{\cos(\theta x+\frac{2m+2}{2}\pi)x^{2m+2}}{(2m+2)!}$$

$$(0<\theta<1)$$

C: $f(x)=\ln(1+x)$

$$f^{(k)}(x)=[\ln(1+x)]^{(k)}=(-1)^{k-1}\frac{(k-1)!}{(1+x)^k}$$

$$f^{(k)}(0)=(-1)^{k-1}(k-1)! \qquad k=1,2,\cdots n$$

则f(x)的n阶麦克劳林公式为

$$\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}+\cdots+(-1)^{n-1}\frac{x^n}{n}+\frac{(-1)^n}{(1+\theta x)^{n+1}}\cdot\frac{x^{n+1}}{n+1}$$

$$(0<\theta<1)$$

D: $f(x)=(1+x)^{\alpha}$

$$f^{(k)}(x)=\alpha(\alpha-1)\cdots(\alpha-k+1)(1+x)^{\alpha-k}$$

$$f^{(k)}(0)=\alpha(\alpha-1)\cdots(\alpha-k+1) \qquad k=1,2,\cdots n$$

则f(x)的n阶麦克劳林公式为

$$(1+x)^{\alpha}=1+\alpha x+\frac{\alpha(\alpha-1)}{2!}x^{2}+\cdots+\frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^{n}$$

$$+\frac{\alpha(\alpha-1)\cdots(\alpha-n)}{(n+1)!}(1+\theta x)^{\alpha-n-1}x^{n+1} \quad (0<\theta<1)$$

常用函数的麦克劳林公式

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + o(x^{2n})$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^n \frac{x^{n+1}}{n+1} + o(x^{n+1})$$

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + o(x^n)$$

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \cdots$$

$$+ \frac{m(m-1)\cdots(m-n+1)}{n!}x^n + o(x^n)$$

计算 $\lim\limits_{x\to 0}\dfrac{e^{x^2}+2\cos x-3}{x^4}$.

解 $\because\quad e^{x^2}=1+x^2+\dfrac{1}{2!}x^4+o(x^4)$

$$\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}+o(x^5)$$

$$\therefore\quad e^{x^2}+2\cos x-3=(\frac{1}{2!}+2\cdot\frac{1}{4!})x^4+o(x^4)$$

$$\text{原式}=\lim_{x\to 0}\frac{\frac{7}{12}x^4+o(x^4)}{x^4}=\frac{7}{12}.$$

证明不等式

设$f''(x)>0$,当$x\to 0$时，$f(x)$与x是等价无穷小，

证明：当$x\neq 0$时，$f(x)>x$

$$f(x)=f(0)+f'(0)x+\frac{f''(\xi)}{2}x^2$$

$$\lim_{x\to 0}\frac{f(x)}{x}=1 \qquad \lim_{x\to 0}f(x)=0=f(0)$$

常用不等式

$$e^x>1+x \quad (x\neq 0) \qquad \sin x<x \quad (0<x\leq \pi)$$

$$\ln(1+x)<x \quad (x>-1, x\neq 0)$$

思考题

利用泰勒公式求极限 $\lim\limits_{x\to 0}\dfrac{e^{x}\sin x - x(1+x)}{x^{3}}$

思考题解答

$$\because e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + o(x^3)$$

$$\sin x = x - \frac{x^3}{3!} + o(x^3)$$

$$\therefore \lim_{x\to 0} \frac{e^x \sin x - x(1+x)}{x^3} =$$

$$\lim_{x\to 0} \frac{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + o(x^3)\right)\left(x - \frac{x^3}{3!} + o(x^3)\right) - x(1+x)}{x^3}$$

$$= \lim_{x\to 0} \frac{\frac{x^3}{2!} - \frac{x^3}{3!} + o(x^3)}{x^3} = \frac{1}{3}.$$

例：求a_0, a_1, a_2, a_3,使当$x \to 1$ 时，有

$$10^x = a_0 + a_1(x-1) + a_2(x-1)^2 + a_3(x-1)^3 + o((x-1)^3)$$

$$a_0 = f(1) = 10, \quad a_1 = f'(1) = 10\ln 10,$$

$$a_2 = \frac{f''(1)}{2!} = 5\ln^2 10, \quad a_3 = \frac{f'''(1)}{3!} = \frac{5}{3}\ln^3 10$$

例：求 $\lim\limits_{x\to 0}\dfrac{\dfrac{x^2}{2}-\sqrt{1+x^2}+1}{x^2\left(\cos x-e^{x^2}\right)}$

$$(1+x^2)^{\alpha}=1+\alpha x^2+\frac{\alpha(\alpha-1)}{2!}x^4+o(x^4)$$

解：原式 $=\lim\limits_{x\to 0}\dfrac{\dfrac{x^2}{2}-\left[1+\dfrac{1}{2}x^2-\dfrac{1}{8}x^4+o\left(x^4\right)\right]+1}{x^2\left\{\left[1-\dfrac{x^2}{2!}+o\left(x^2\right)\right]-\left[1+x^2+o\left(x^2\right)\right]\right\}}$

$$=\lim_{x\to 0}\frac{\dfrac{1}{8}x^4+o\left(x^4\right)}{x^2\left\{-\dfrac{3}{2}x^2+o\left(x^2\right)\right\}}=-\frac{1}{12}$$

例: $\lim\limits_{x\to 0}\left(\dfrac{a^x - x\ln a}{b^x - x\ln b}\right)^{\frac{1}{x^2}}$ $\quad (a, b > 0, a \neq 1, b \neq 1, a \neq b)$

$$a^x = e^{\ln a^x} = e^{x\ln a}$$

解: $a^x = 1 + x\ln a + \dfrac{x^2}{2!}\ln^2 a + o(x^2)$, $\quad b^x = 1 + x\ln b + \dfrac{x^2}{2!}\ln^2 b + o(x^2)$,

$$\ln t = \ln(1 + (t-1)) \sim t - 1 \qquad t \to 1$$

$$\text{原式} = \exp\lim_{x\to 0}\left[\frac{1}{x^2}\left(\frac{1 + x\ln a + \frac{x^2}{2}\ln^2 a + o(x^2) - x\ln a}{1 + x\ln b + \frac{x^2}{2}\ln^2 b + o(x^2) - x\ln b} - 1\right)\right]$$

$$= \exp\left[\lim_{x\to 0}\frac{1}{x^2}\cdot\frac{\frac{x^2}{2}(\ln^2 a - \ln^2 b) + o(x^2)}{1 + x\ln b + \frac{x^2}{2}\ln^2 b + o(x^2)}\right] = e^{\frac{1}{2}\left[\ln^2 a - \ln^2 b\right]}$$

例：求 $\lim\limits_{x\to\infty} x[2x-1-2x^2\ln(1+\frac{1}{x})]$

解：令 $x=\frac{1}{t}$ 由 $\ln(1+t)=t-\frac{1}{2}t^2+\frac{1}{3}t^3+o(t^3)$

$$原式=\lim_{t\to 0}\frac{1}{t}\left[\frac{2}{t}-1-\frac{2}{t^2}\ln\left(1+t\right)\right]$$

$$=\lim_{t\to 0}\frac{2t-t^2-2[t-\frac{1}{2}t^2+\frac{1}{3}t^3+o(t^3)]}{t^3}$$

$$=-\frac{2}{3}$$

求极限

$$\lim_{x\to\infty}[ax+(a^2-x^2)\ln(1+\frac{a}{x})]$$

$$\overset{x=\frac{1}{t}}{=}\lim_{t\to 0}[\frac{a}{t}+(a^2-\frac{1}{t^2})\ln(1+at)]$$

$$=\lim_{t\to 0}\frac{at+(at^2-1)\ln(1+at)}{t^2}$$

$$=\lim_{t\to 0}\frac{a+2at\ln(1+at)+\frac{a(at^2-1)}{1+at}}{2t}$$

$$=\lim_{t\to 0}\frac{a^2t(1+t)}{2t(1+at)}+\lim_{t\to 0}\frac{2at\ln(1+at)}{2t}=\frac{a^2}{2}$$

$$\lim_{x\to\infty}[ax+(a^2-x^2)\ln(1+\frac{a}{x})]$$

$$=\lim_{x\to\infty}[ax+(a^2-x^2)(\frac{a}{x}-\frac{a^2}{2x^2}+o(\frac{1}{x^2}))]$$

$$=\lim_{x\to\infty}[ax+\frac{a^3}{x}-ax-\frac{a^4}{2x^2}+\frac{a^2}{2}+o(\frac{1}{x^2})+\frac{o(\frac{1}{x^2})}{\frac{1}{x^2}}]$$

$$=\frac{a^2}{2}$$

例：设f″(x)在x=a连续，f(a)=0,f′(a) ≠ 0.

求 $\lim\limits_{x\to a}\left[\dfrac{f(x)}{f'(a)(x-a)}\right]^{\frac{1}{x-a}}$

解：原式$=\lim\limits_{x\to a}\left[\dfrac{f(a)+f'(a)(x-a)+\frac{1}{2}f''(a)(x-a)^2+o((x-a)^2)}{f'(a)(x-a)}\right]^{\frac{1}{x-a}}$

$$=\lim_{x\to 0}\left[1+\frac{f''(a)}{2f'(a)}(x-a)+o(x-a)\right]^{\frac{1}{x-a}}$$

$\ln(1+x)\sim x \qquad (x\to 0)$

$$\overset{1^\infty}{=}\exp\left[\lim_{x\to 0}\frac{\frac{f''(a)}{2f'(a)}(x-a)+o(x-a)}{x-a}\right]=e^{\frac{f''(a)}{2f'(a)}}$$

设**f(x)**在**[a,b]**上连续，在**(a,b)**内二阶连续可导，试证明存在 $\xi \in (a,b)$ ，使 $f(b)-2f(\frac{a+b}{2})+f(a)=\frac{(b-a)^2}{4}f''(\xi)$

$$f(a)=f(\frac{a+b}{2})+f'(\frac{a+b}{2})(a-\frac{a+b}{2})+\frac{f''(\xi_1)}{2}(a-\frac{a+b}{2})^2$$

$$f(b)=f(\frac{a+b}{2})+f'(\frac{a+b}{2})(b-\frac{a+b}{2})+\frac{f''(\xi_2)}{2}(b-\frac{a+b}{2})^2$$

$$f(b)+f(a)-2f(\frac{a+b}{2})=\frac{(b-a)^2}{4}[\frac{f''(\xi_1)+f''(\xi_2)}{2}]$$

$f''(x)$ 在 $[\xi_1,\xi_2]$ 上连续，故存在最大值 M 和最小值 m

$$m \le \frac{1}{2}[f''(\xi_1)+f''(\xi_2)] \le M$$

$$\exists \xi \in [\xi_1,\xi_2] \subset (a,b), \quad f''(\xi)=\frac{1}{2}[f''(\xi_1)+f''(\xi_2)]$$

设$f(x)$在$[0,2]$上连续，$(0,2)$内有三阶导数，且

$$f(0)=3,\ f(2)=4,\quad f(1)=\max_{x\in[0,2]} f(x)$$

试证明：$\exists\xi\in(0,2)$，使 $f'''(\xi)\geq 3.$

证明：由$f(1)=\max\limits_{x\in[0,2]} f(x)$，知$x=1$为极大点，

故$f'(1)=0$　　将$f(0), f(2)$在$x=1$处展开，有

$$f(0)=f(1)+f'(1)(0-1)+\frac{f''(1)}{2!}(0-1)^2+\frac{f'''(\xi_1)}{3!}(0-1)^3, 0<\xi_1<1$$

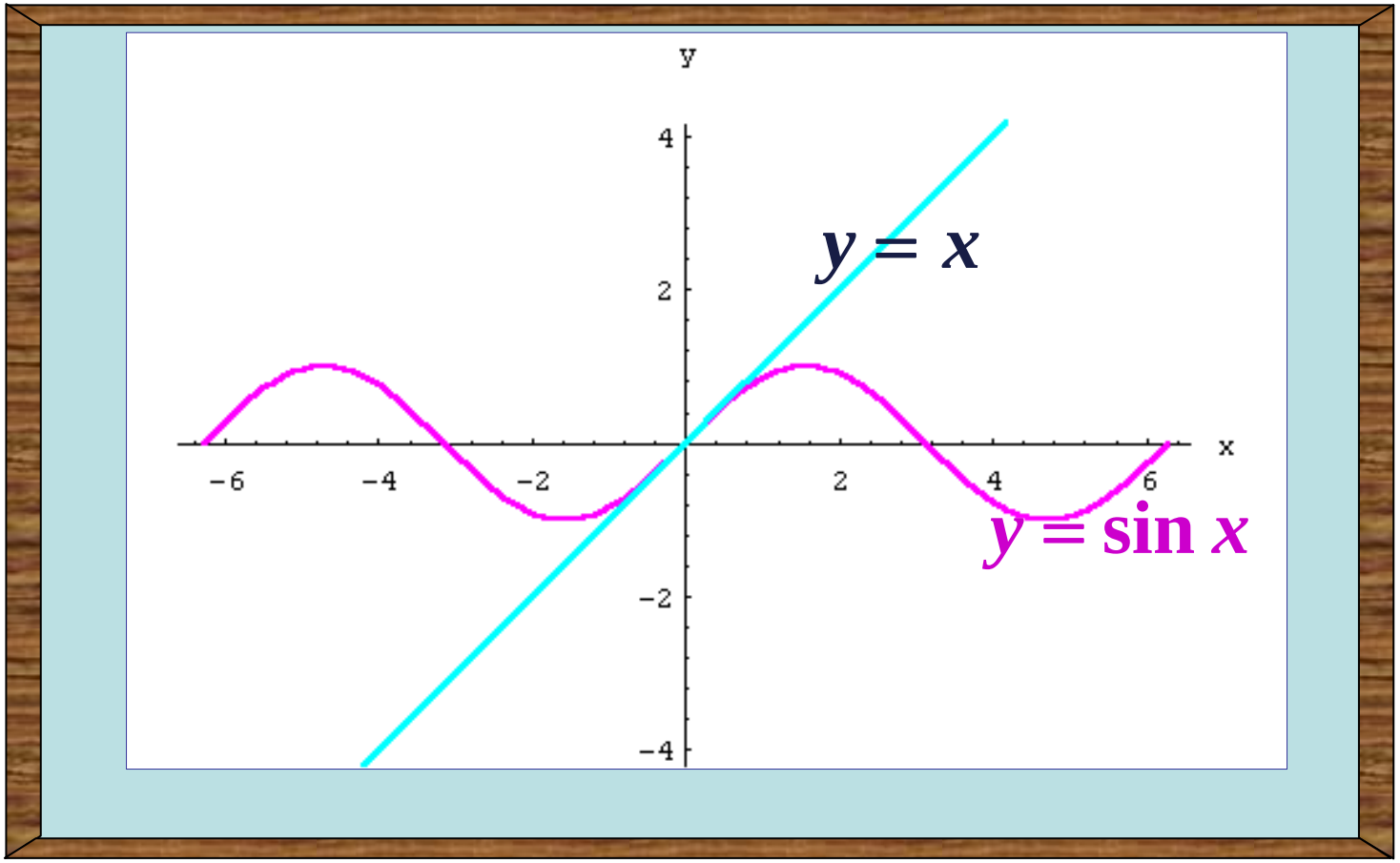
$$f(2)=f(1)+f'(1)(2-1)+\frac{f''(1)}{2!}(2-1)^2+\frac{f'''(\xi_2)}{3!}(2-1)^3, 1<\xi_2<2$$

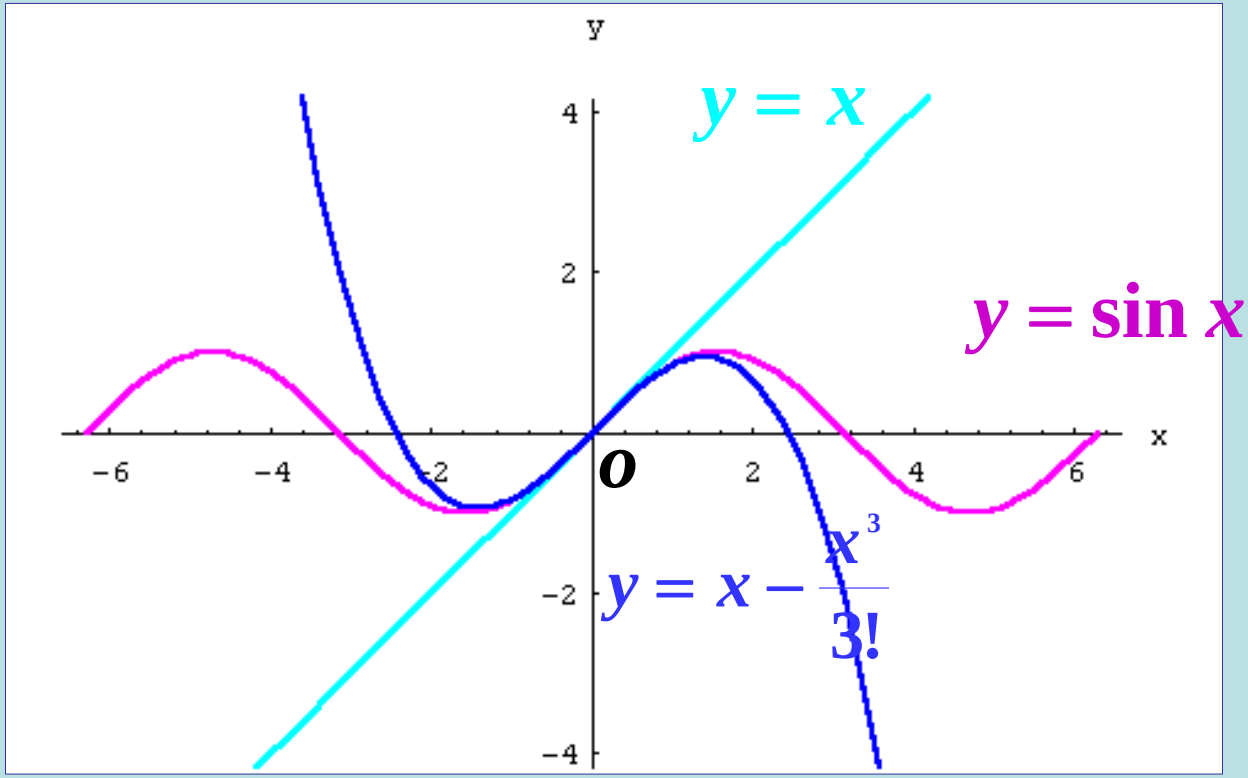
两式相减，　$f'''(\xi_1)+f'''(\xi_2)=6$

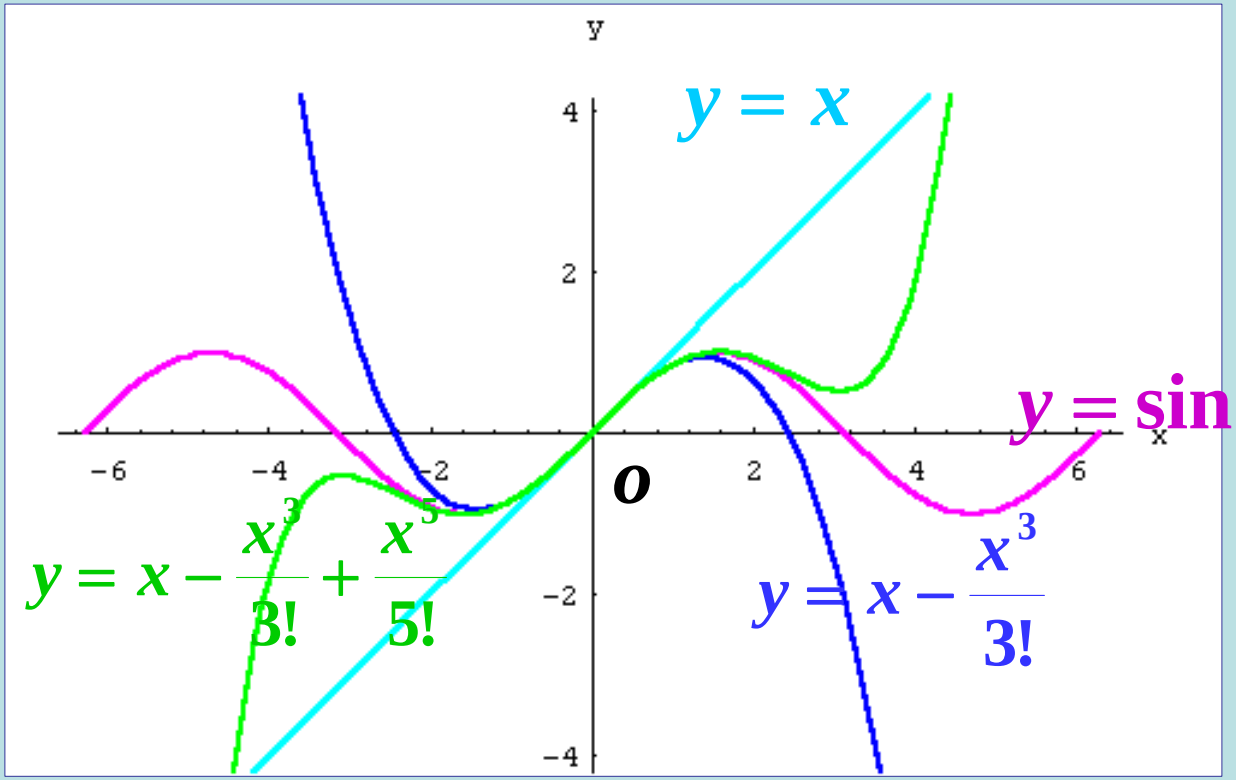
这两项中至少有一项不 小于3. 即结论成立！

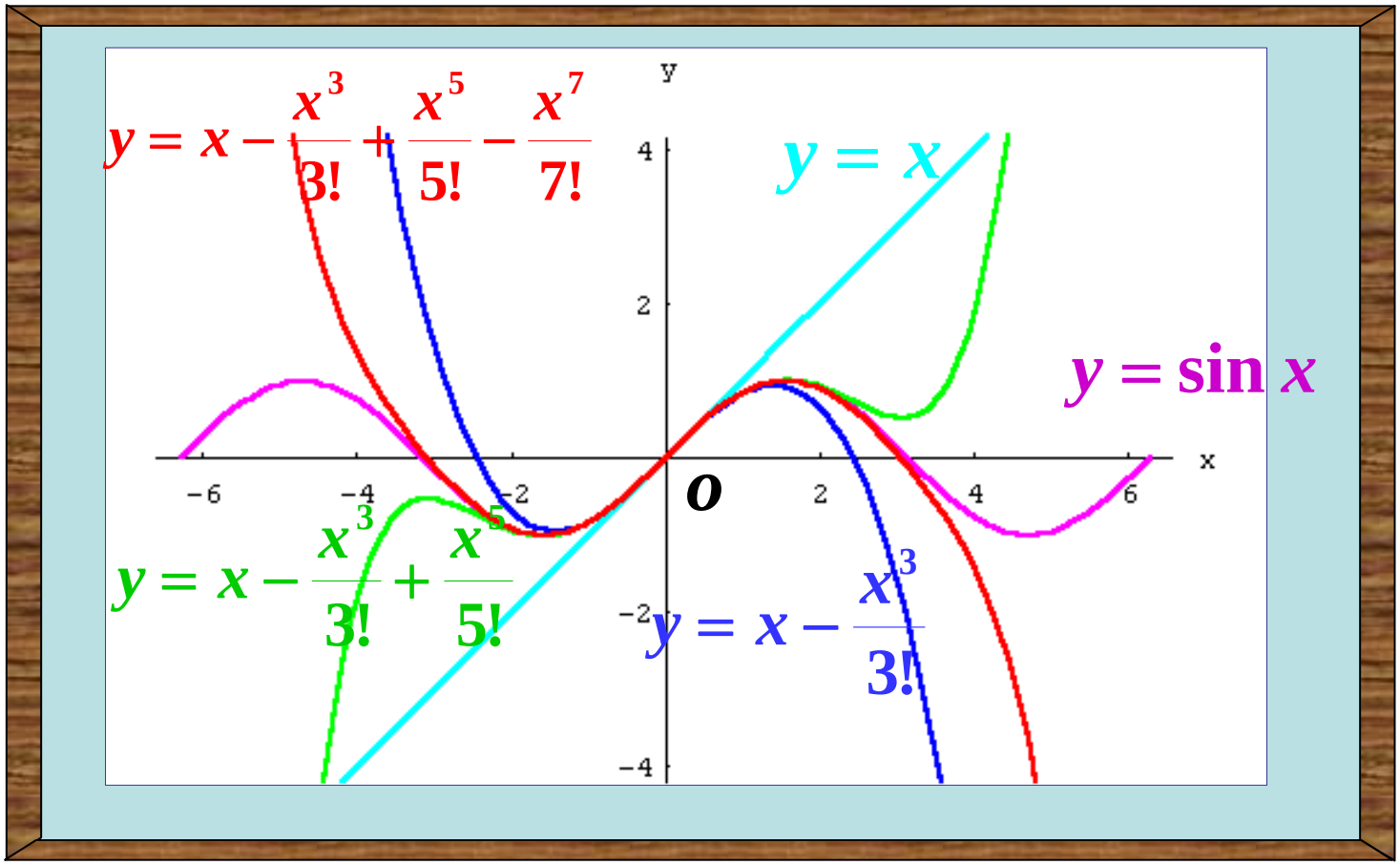
五、小结

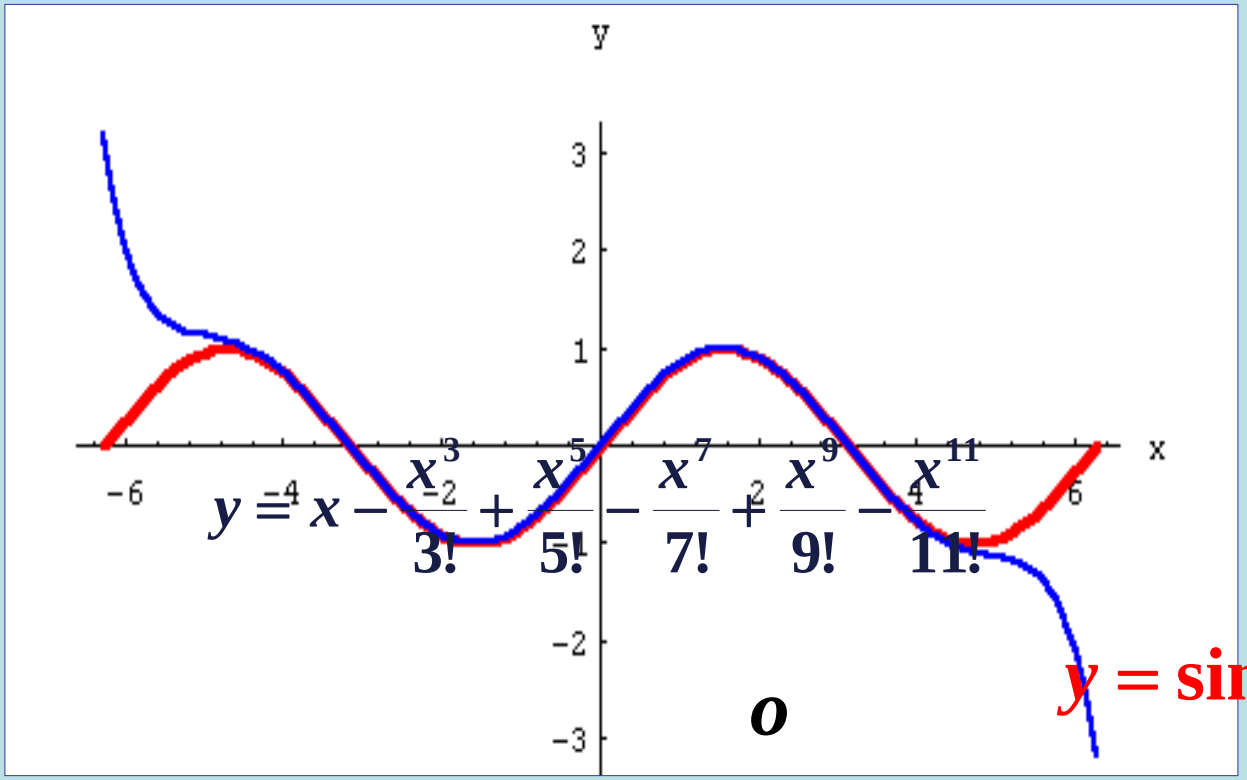
1. Taylor 公式在近似计算中的应用；




y
x
4
2
-2
-4
-6
-4
-2
2
4
6
o
$y = x$
$y = \sin x$
$y = x - \frac{x^3}{3!}$


y
x
4
2
-2
-4
-6
-4
-2
2
4
6
o
$y = x$
$y = \sin x$
$y = x - \frac{x^3}{3!} + \frac{x^5}{5!}$
$y = x - \frac{x^3}{3!}$


$y = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$
$y = x$
$y = \sin x$
$y = x - \frac{x^3}{3!} + \frac{x^5}{5!}$
$y = x - \frac{x^3}{3!}$
y
x
o
4
2
-2
-4
-6
-4
-2
2
4
6



$$y = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!}$$

o

$y = \sin x$